A Novel Spatio-Temporal Rate-Splitting-Based Power Allocation Optimization Strategy for RIS-Assisted 6G MU-MISO Communication Systems

Xuejie Hu, *Student Member, IEEE*, Yue Tian, *Member, IEEE*, Zhidu Li, *Senior Member, IEEE*, Xianling Wang, *Member, IEEE*, Baiyun Xiao, *Member, IEEE*, Yau Hee Kho, *Senior Member, IEEE*, and Wenda Li, *Member, IEEE*

***Abstract*—Reconfigurable intelligent surfaces (RISs) can dynamically adjust phase shifts to achieve scalability in networks, while rate-splitting multiple access (RSMA) can dynamically allocate spectrum and power resources according to the communication needs of IoT devices and reduce the energy consumption of the system. Therefore, the integration of RIS and RSMA can not only enhance the performance of IoT communication systems, but also reduces wastage of limited resources. This paper proposes a novel spatio-temporal rate-splitting-based power allocation optimization strategy for RIS-assisted multi-user (MU) systems to maximize channel capacity and energy efficiency gains. Leveraging the proposed spatio-temporal minimum mean squared error (STMMSE) principle and the rate splitting-based optimal power consumption (RS-OPC) method, the approach considers the Cramér-Rao bound for channel errors and derives expressions for maximizing channel capacity and obtaining the optimal solution. The proposed method selectively adjusts power values for different users and transmission types of symbols to achieve optimal power allocation objectives, thereby ensuring communication quality while optimizing channel capacity. This offers communication systems a higher configurability and resource optimization. Extensive simulation results including channel capacity, energy efficiency (EE) and spectral efficiency (SE) validate the effectiveness of the proposed method.**

***Index Terms*—Reconfigurable intelligent surfaces (RISs), NOMA, MU-MISO, power consumption optimization, source allocation.**

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Xuejie Hu, Yue Tian, and Xianling Wang are with the Fujian Key Laboratory of Communication Network and Information Processing, Xiamen University of Technology, Xiamen 361024, China (e-mail: huxjhk@outlook.com; yue.tian.xmut@outlook.com; xianling.wang@ieee.org).

Zhidu Li is with the School of Communication and Information Engineering, Chongqing University of Posts and Telecommunications, Chongqing 400065, China (e-mail: lizd@cqupt.edu.cn).

Baiyun Xiao and Yau Hee Kho are with the School of Engineering and Computer Science, Victoria University of Wellington, Wellington 6012, New Zealand (e-mail: xxbynl@outlook.com; yauhee.kho@vuw.ac.nz).

Wenda Li is with the School of Engineering and Physical Sciences, Heriot-Watt University, EH14 4AS Edinburgh, U.K. (e-mail: Wenda.Li@hw.ac.uk).

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I. Introduction

RECONFIGURABLE intelligent surfaces (RISs) are a promising technology for improving spectrum and energy efficiency in sixth-generation (6G) wireless communication systems, owing to their energy-saving, easy-to-deploy, and cost-effective characteristics [1]. Deploying RISs in Internet of Things (IoT) systems can extend communication links, enhance the reliability of millimeter-wave communication systems [2], and improve the security of wireless links [3] as well as the positioning accuracy [4]. Furthermore, the robust support provided by digital twin (DT) technology [5] contributes to the development and deployment of RISs, addressing the ever-growing communication demands and challenges [6], [7]. Rate-splitting multiple access (RSMA) allows multiple users to share the same spectrum resources simultaneously [8]. Its flexibility enables the dynamic allocation of resources based on users' communication needs, channel conditions, and network load, thereby maximizing transmission reliability and reducing latency in emergency communication [9], [10].

Adopting RSMA in the design of RIS-assisted wireless communication systems is crucial for future IoT networks. It not only leverages rate splitting to protect user privacy [11] and secure sensitive communication information but also enhances the energy efficiency of wireless communication [12], [13]. The integration of RIS and RSMA technologies effectively extends the operational lifetime of energy-constrained, low-power networks, thereby reducing operational costs. Moreover, the use of deep reinforcement learning (DRL) algorithms combined with proximal policy optimization has the potential to significantly improve the rate and user fairness in RIS-assisted RSMA systems [14], [15]. This holistic approach addresses energy efficiency and performance optimization challenges, making RIS-assisted RSMA systems a promising solution for future IoT networks.

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RIS-assisted RSMA systems exhibit notable advantages in interference management and communication performance enhancement [16]. By leveraging dynamic beamforming and intelligent reflection, they achieve efficient multi-user interference isolation [17]. Power allocation is a core aspect of optimization in RIS-assisted RSMA systems. Properly allocating power between the common and private streams, combined with real-time adjustments to dynamic environments, can further improve system performance and enable energy-efficient communication services. However, due to the complexity of multi-user channel conditions and the constraints on computational resources [18], [19], RIS-assisted RSMA power allocation faces multiple challenges, including adaptive algorithm design, multi-user fairness, and the coupling of joint RIS configuration and power allocation optimization. Developing efficient dynamic power allocation algorithms and multi-objective optimization frameworks requires further research and refinement.

A. Related Work

In pursuit of higher spectral efficiency, the joint optimization problem of RIS-assisted wireless communication networks has been subject to further research [20], [21], [22]. [23] proposed the use of projection onto the gradient ascent method (PGAM), relying on channel state information to maximize the sum spectral efficiency (SE) of complex systems. This approach boasted low complexity and minimal overhead. To enhance the security of space-ground communication systems, [24] introduced a joint optimization method for ground station beamformers and two different RIS reflection vectors in a dual RIS scenario. Additionally, for a single RIS, an alternating optimization (AO) algorithm was employed to address the problem of maximizing secrecy rate. [25] utilized a one-dimensional (1D) grid search method to approximate optimal solutions for the asymptotic problem, providing an effective approach for fundamental performance-complexity trade-offs in RIS partition design. Addressing the rate optimization problem in RIS-assisted mixed non-orthogonal multiple access wireless networks, an AO framework, along with an alternating direction method [26] combining norm approximation, max-min optimization, and multiplier [27], effectively tackled time and power allocation problems for active and passive beamforming vectors. Considering the imperfect and outdated channel state information for the autonomous aerial vehicle (AAV)-RIS-user composite channel, a capacity maximization strategy based on RIS-AAV non-orthogonal multiple access (NOMA) networks was investigated in [28] to optimize the AAV trajectory, while [29] proposed an approach analyzing network performance in terms of average coverage probability (ACP) and average bit error rate (ABER) over the entire AAV flight time, providing insights into RIS selection in such scenarios.

RSMA, as a multi-access technology developed on the foundation of traditional multiple access techniques, addresses the issue of multi-user interference in multiple input multiple output (MIMO) systems [30], [31], [32]. In a network involving an RIS-assisted base station and multiple users served by a hybrid aerial full-duplex autonomous aerial vehicle relay, [33] investigated the joint optimization of RSMA parameters, RIS three-dimensional coordinates, and the phase-shift matrix of RIS. The study employed the block coordinate descent (BCD) method to obtain the optimal position of the autonomous aerial vehicle and network performance. For the cognitive RSMA (C-RSMA) system, [34] analyzed the optimal transmission rate schemes and the order of interference cancellation in signal decoding sequence to assist users in achieving the maximum achievable rate. [35] proposed a DRL-based deep deterministic policy gradient (DDPG) approach to address the problem of designing the motion trajectory of autonomous aerial vehicles in the downlink RSMA system without prior knowledge of the dynamic environment. In the context of short packet transmission in 5-th generation (5 G) networks, data-oriented approaches introduced instantaneous performance metrics for small data transmissions. [36] presented an optimization method for data-oriented RSMA downlink systems, providing an efficient design for multi-user short packet transmission. A hybrid data-driven deep learning (DL)-based RSMA precoding scheme [37] addressed the challenge of obtaining accurate high-dimensional channel state information under limited pilot and feedback signal overhead. [38] used particle swarm optimization and semidefinite relaxation methods to solve the power minimization problem in the RSMA framework of a multiple-input single-output (MISO) cognitive radio system.

The combination of RSMA and RIS, as a novel technology to enhance the energy and spectrum efficiency of multi-antenna networks, has the potential to significantly reduce transmission power by establishing optimized strategies [39], [40], [41]. Various studies have addressed different non-convex optimization problems using methods such as successive convex approximation (SCA), exterior-point methods, AO techniques [42], [43], jointly optimizing precoding vectors, beamforming, and common stream splitting. The research results have indicated a significant progress in minimizing the overall energy consumption. In the design problem of resource allocation in space-time adaptive radio (STAR)-assisted RSMA systems, [44] proposed a mixed-integer nonlinear programming method for the joint optimization of power allocation and decoding order to maximize the system's total throughput. [45] presented a max-min fair transmission scheme for RIS-assisted RSMA multigroup multicasting, addressing the issue of fair and high-quality multi-user services under spectrum constraints. [46] demonstrated, through constrained problems, that in beyond diagonal (BD) RIS-assisted ultra-reliable low-latency communication systems, RSMA can greatly enhance system performance compared to conventional RIS. To effectively mitigate high node interference in multi-user scenarios, [47] proposed a low-complexity user pairing scheme based on K-means clustering, considering the total throughput maximization problem under power allocation optimization for uplink RSMA systems. [48] explored an unconventional ultra-reliable low latency communication (xURLLC) service in RIS-assisted RSMA systems under finite block-length (FBL) transmission constraints to promote mobile broadband reliable low latency communication (mBRLLC). The study utilized an AO approach to solve each sub-optimization problem proposed.

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B. Motivation and Contributions

In the multi-user RIS-assisted RSMA system, the varying activity levels of different users necessitate optimized time and power allocation to maximize the channel capacity for all users, thereby enhancing user satisfaction. In this process, accurate channel estimation and optimal resource allocation are critical, as the precision of channel information directly affects the effectiveness of time and power resource allocation. Effective resource allocation can significantly improve the overall energy efficiency of the system. The transmission power of training symbols directly impacts the quality of channel estimation, while optimizing the transmission power of data symbols can further enhance the energy efficiency of system transmissions. Additionally, optimizing the time allocation factor based on the channel conditions of different users and the utilization of channel resources can achieve a more balanced time allocation, thereby increasing the total channel capacity. Based on this, this paper proposes the spatio-temporal minimum mean squared error (STMMSE) principle, which leverages the channel's channel estimation error (CEE) to derive channel capacity and optimizes power and time allocation, considering the contributions of training power, data power, and time to channel capacity. Furthermore, to optimize power allocation during the data transmission phase and maximize total channel capacity, a rate splitting-based optimal power consumption (RS-OPC) method is proposed, providing an extended solution for multi-user fairness in resource allocation in wireless networks.

Building upon prior research efforts, the focus of our investigation lies in the optimization of total channel capacity in the context of multi-user scenarios utilizing RISs-assisted RSMA (R-RS) technology. In this setup, the base station (BS) communicates with both proximal and distant users through RISs. Communication data is categorized into common streams and private streams, with different transmission rate assigned to distinct types of communication data. The primary contributions of this paper are as follows:

- Proposes a novel method to maximize the total channel capacity of a multi-user (MU)-MISO system using RISs-assisted RSMA technology. Each UE is served by a BS employing RISs. Specifically, the optimization problem under threshold achievable rate, total transmission time, and total power constraints is investigated. The non-convex problem is decomposed into two sub-problems, addressing the optimization of the user transmission time allocation coefficient matrix and power allocation, respectively.
- For the optimization of the user transmission time allocation coefficient matrix, the STMMSE principle is employed, taking into account the differences in channel characteristics and transmission requirements among users. By optimizing the appropriate transmission time allocated to each user, the communication resources are effectively utilized to enhance the overall channel capacity. The use of STMMSE improves user fairness in the communication system, and the user transmission time allocation coefficient matrix can be dynamically adjusted based on varying communication needs and network conditions, allowing the system to maintain high performance under different load conditions.
- For different types of transmitted data, the data is categorized into training symbols and data symbols. The transmission power is optimized for different symbol types, utilizing the STMMSE principle. The Cramér-Rao lower bound (CRLB) for channel estimation error is derived using Lagrange multiplier method. Finally, the principle of optimal power allocation to obtain optimal allocation coefficients is employed to training and data symbols, aiming to maximize the overall capacity.
- Building upon the optimized maximization of channel capacity and optimal power allocation employed to training and data symbols, the RS-OPC method is employed to determine the optimal transmission power for proximal and distant users and different types of data, including both common and private streams for data symbols. This ensures that, while simultaneously transmitting different data types, the system efficiency is maximized to the greatest extent possible.

C. Organizations and Notations

The rest of the paper is organized as follows. Section II describes the system model of R-RS system. Section III contains the proposed solutions and algorithms for the formulated optimization problem. Simulation results and discussions of the proposed algorithm adapted to R-RS system are represented in Section IV. Finally, Section V shows the conclusion of the paper.

Notations: $\mathbf{I}$ means the identify matrix, $\mathbb{E}(\cdot)$ is expectation. $(\cdot)^H$, $(\cdot)^T$, $(\cdot)^*$ and $(\cdot)^{-1}$ express conjugate transpose, transpose, conjugate and inverse, respectively. $\|\cdot\|$ indicates the Euclidean norm. $\det(\cdot)$ and $\mathrm{Tr}(\cdot)$ represent the determinant and trace of a matrix. $\mathbb{C}^{N\times N}$ stands for the set of real (complex) matrices of size $N \times N$. The distribution of the circularly symmetric complex Gaussian (CSCG) is denoted as $n \sim \mathcal{CN}(0, \sigma^2)$ with zero mean and variance σ^2. $\mathrm{diag}(\mathrm{A})$ describes the diagonal matrix with vector A for its diagonal elements, while arg(A) is the argument of the complex number A.

II. System Model of Spatio-Temporal RIS-Assisted RSMA Network

A. System Model

In the proposed scenario, a BS with the number of N transmitting antennas and multiple RISs are considered to serve UEs with single antenna together, as shown in Fig. 1 with the proximal users set denoted as $\mathcal{K} = \{1, \ldots, k, \ldots, K\}$ and the distant users set denoted as $\mathcal{L} = \{1, \ldots, l, \ldots, L\}$. There are Z elements on each RIS. The set of RISs is represented as $\mathcal{V} = \{1, \ldots, v, \ldots, V\}$, the collection of elements on RIS is represented as $\mathcal{Z} = \{1, \ldots, z, \ldots, Z\}$, the UE collection is represented as $\mathcal{U} = \{1, \ldots, u, \ldots, U\}$. In this model, the data structure consisting of $M_{u,t}$ training symbols S_t and $M_{u,d}$ data symbols S_d for each spatial stream is considered. Define $\mathbf{S} = [S_t, S_d]$, where $S_t = [S_t(1), \ldots, S_t(N)]^T \in \mathbb{C}^{N\times M_{u,t}}$

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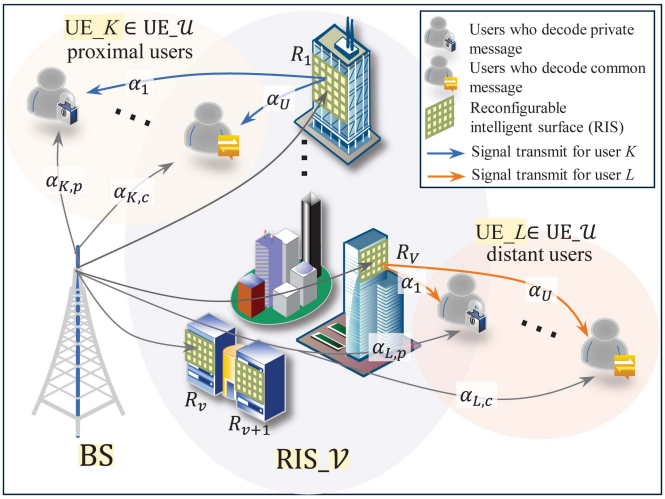

Fig. 1. System model for the network.

and $S_d = [S_d(1), \ldots, S_d(N)]^T \in \mathbb{C}^{N\times M_{u,d}}$. Let P be the total power of the transmit symbols, p_t the power of training symbols, and p_d the power of the data symbols, where $P = p_t + p_d$. The average power of the training and data symbols are $p^2_{t,ave} = \frac{1}{NM_{u,t}}\sum_{n=1}^{N} S_t(n)^H S_t(n)$ and $p^2_{d,ave} = \frac{1}{NM_{u,d}}\sum_{n=1}^{N}\mathbb{E}[S_d(n)^H S_d(n)]$, where $S_t(n)$ and $S_d(n)$ denote the training and data symbols transmitted from the n-th antenna, respectively.

$\mathbf{h}_k \in \mathbb{C}^{1\times N}$ and $\mathbf{h}_l \in \mathbb{C}^{1\times N}$ are denoted as the direct channel matrix from the BS to k-th proximal and l-th distant user. $\tilde{\mathbf{H}}_{vz} \in \mathbb{C}^{Z\times N}$ and $\tilde{\mathbf{h}}^k_{vz} \in \mathbb{C}^{1\times Z}$ represent the channel matrix from the BS to the Z reflect element of v-th RIS and the channel matrix from the Z reflect element of v-th RIS to the k-th user, $\tilde{\mathbf{h}}^l_{vz} \in \mathbb{C}^{1\times Z}$ indicates the channel matrix from the Z reflect element of v-th RIS to l-th user, where $\tilde{\mathbf{H}}_{vz} = \mathbf{H}_{vz}e^{-j\varphi_z}$, $\tilde{\mathbf{h}}^k_{vz} = \mathbf{h}^k_{vz}e^{-j\partial_{K,z}}$, $\tilde{\mathbf{h}}^l_{vz} = \mathbf{h}^l_{vz}e^{-j\partial_{L,z}}$. $\mathbf{H}_{vz}$, $\mathbf{h}^k_{vz}$ and $\mathbf{h}^l_{vz}$ represent the channel coefficient matrix representing the BS to the Z reflect element of v-th RIS, the channel coefficient matrix representing the Z reflect element of v-th RIS to proximal user, and the channel coefficient matrix representing the Z reflect element of v-th RIS to distant user, respectively. φ_z indicates the phase of $\tilde{\mathbf{H}}_{vz}$, while $\partial_{K,z}$ and $\partial_{L,z}$ express the phases of $\tilde{\mathbf{h}}^k_{vz}$ and $\tilde{\mathbf{h}}^l_{vz}$. The transmit signal to proximal and distant user is denoted by $\mathbf{s}_K = [s_{K,p}, s_{K,c}]^T$ and $\mathbf{s}_L = [s_{L,p}, s_{L,c}]^T$, where $s_{K,p}$ and $s_{K,c}$ is the private message and common message of proximal user, $s_{L,p}$ and $s_{L,c}$ express the private message and common message of distant user. The signal received by proximal and distant user is represented as:

$$\begin{aligned}\mathbf{y}_K &= \prod_{v=1}^{V}\left(\mathbf{h}_k + \tilde{\mathbf{h}}^k_{vz}\Theta_{v,z}\tilde{\mathbf{H}}_{vz}\right)\sqrt{p_K}\cdot\mathbf{s}_K + n_K \\ &= \sum_{*=p,c}\prod_{v=1}^{V}\left(\mathbf{h}_k + \tilde{\mathbf{h}}^k_{vz}\Theta_{v,z}\tilde{\mathbf{H}}_{vz}\right) \\ &\quad\times\sqrt{\alpha_{K,*}\cdot p_K}\cdot s_{K,*} + n_K, \end{aligned} \tag{1}$$

$$\begin{aligned}\mathbf{y}_L &= \prod_{v=1}^{V}(\mathbf{h}_l + \tilde{\mathbf{h}}^l_{vz}\Theta_{v,z}\tilde{\mathbf{H}}_{vz})\sqrt{p_L}\cdot\mathbf{s}_L + n_L \\ &= \sum_{*=p,c}\prod_{v=1}^{V}(\mathbf{h}_l + \tilde{\mathbf{h}}^l_{vz}\Theta_{v,z}\tilde{\mathbf{H}}_{vz}) \\ &\quad\times\sqrt{\alpha_{L,*}\cdot p_L}\cdot s_{L,*} + n_L. \end{aligned} \tag{2}$$

where $\Theta_{v,z} = \mathrm{diag}[g_{1,z}\phi(\theta_{1,z}), g_{2,z}\phi(\theta_{2,z}), \ldots, g_{v,z}\phi(\theta_{v,z})]$ represents the phase-shift matrix of RIS, and $\prod_{v=1}^{V}$ indicates that the signal transmission considers the combined effects of multiple RISs, with each RIS indexed by v. $\phi(\theta_{v,z}) = e^{j\theta_{v,z}}$, $g_{v,z} \in (0,1)$ is amplitude reflection coefficient, $\theta_{v,z} \in [0, 2\pi]$ stands for phase-shift of i-th RIS. n_K, $n_L \sim \mathcal{CN}(0, \sigma^2_n)$ represents the Gaussian white noise of the channel with a mean of zero and a variance of σ^2_n. $\alpha_{K,p}$ and $\alpha_{K,c}$ are the power coefficients allocated to $S_{K,p}$ and $S_{K,c}$, $\alpha_{L,p}$ and $\alpha_{L,c}$ are the power coefficients allocated to $S_{L,p}$ and $S_{L,c}$. The transmit power from BS to proximal and distant users are denoted by p_K and p_L. Let $\mathbf{h}_l$, $\mathbf{h}_k$, $\mathbf{h}^l_{vz}$, $\mathbf{h}^k_{vz}$ and $\mathbf{H}_{vz}$ be random variables following the Rayleigh distribution with their PDF and CDF, parameterized by $\beta_{\mathbf{H}}$, expressed as:

$$f(x) = \frac{x}{\beta_{\mathbf{H}}}\exp\left(-\frac{x^2}{2\beta_{\mathbf{H}}}\right), F(x) = 1 - \exp\left(-\frac{x^2}{2\beta_{\mathbf{H}}}\right), x \geq 0, \tag{3}$$

where $\beta_{\mathbf{H}} = \mathbb{E}[x^2]/2$ indicates the scale parameter of the distribution. The mean and variance of the distribution are expressed as $\sqrt{\frac{\beta_{\mathbf{H}}}{2}\pi}$ and $(2 - \frac{\pi}{2})\beta_{\mathbf{H}}$.

B. Channel Estimation Model

For brevity, let $\mathbf{H}_{RIS} = \sum_{\Delta=k,l}\mathbf{h}_k + \tilde{\mathbf{h}}^k_{vz}\Theta_{v,z}\tilde{\mathbf{H}}_{vz}$. Define $\hat{\mathbf{H}}_{RIS}$ and $\Delta\mathbf{H}_{RIS}$ as the estimated channel and the estimation error, assumed to be uncorrelated. The received data signals can be given by:

$$\begin{aligned} Y_d &= \mathbf{H}_{RIS}\mathbf{S}_d + N_d = (\hat{\mathbf{H}}_{RIS} + \Delta\mathbf{H}_{RIS})\mathbf{S}_d + N_d \\ &= \hat{\mathbf{H}}_{RIS}\mathbf{S}_d + \underbrace{\Delta\mathbf{H}_{RIS}\mathbf{S}_d + N_d}_{N'_d}, \end{aligned} \tag{4}$$

where $\mathbf{H}_{RIS} = [\mathbf{H}_{1,RIS}, \ldots, \mathbf{H}_{j,RIS}, \ldots, \mathbf{H}_{N,RIS}] \in \mathbb{C}^{N\times N}$ and $\Delta\mathbf{H}_{RIS} = [\Delta\mathbf{H}_{1,RIS}, \ldots, \Delta\mathbf{H}_{j,RIS}, \ldots, \Delta\mathbf{H}_{N,RIS}] \in \mathbb{C}^{N\times N}$ is the channel matrix and channel estimation error matrix with j-th row. $\mathbf{H}_{j,RIS}$ and $\Delta\mathbf{H}_{j,RIS}$ are denoted as:

$$\mathbf{H}_{j,RIS} = \sum_{\star=k,l}(\mathbf{h}_{j,\star} + \tilde{\mathbf{h}}^{\star}_{j,vz}\Theta_{v,z}\tilde{\mathbf{H}}_{j,vz}), \tag{5}$$

$$\Delta\mathbf{H}_{j,RIS} = \sum_{\star=k,l}(\Delta\mathbf{h}_{j,\star} + \Delta\tilde{\mathbf{h}}^{\star}_{j,vz}\Theta_{v,z}\Delta\tilde{\mathbf{H}}_{j,vz}), \tag{6}$$

where $\Delta\mathbf{h}_{j,k}$, $\Delta\mathbf{h}_{j,l}$ and $\Delta\tilde{\mathbf{H}}_{j,vz}$ are the estimated channel matrix from BS to k-th proximal, from BS to l-th distant user and from BS to the z-th reflect element of v-th RIS. $\Delta\tilde{\mathbf{h}}^k_{j,vz}$ and $\Delta\tilde{\mathbf{h}}^l_{j,vz}$ represent the estimated channel matrix from the z-th reflect element of v-th RIS to the k-th user and l-th user, respectively.

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C. Rate-Splitting Model

Due to the transmission time for the proximal user and the distant user may not be equal, the time fractions assigned to proximal user and distant user will be denoted as T_{τ_K} and T_{τ_L}, respectively. The received SINR and the corresponding achievable rate at the proximal user (decoding common message) is given by:

$$\Upsilon_{K,c} = \frac{\left|(\mathbf{h}_k + \tilde{\mathbf{h}}_{vz}^k \Theta_{v,z} \tilde{\mathbf{H}}_{vz})\sqrt{\alpha_{K,c} p_K}\right|^2}{\left|(\mathbf{h}_k + \tilde{\mathbf{h}}_{vz}^k \Theta_{v,z} \tilde{\mathbf{H}}_{vz})\sqrt{\alpha_{K,p} p_K}\right|^2 + \sigma_n^2}, \quad (7)$$

$$R_{K,c}[\Upsilon_{K,c}] = T_{\tau_K} \log_2(1 + \Upsilon_{K,c}), \quad (8)$$

while the received SINR and the corresponding achievable rate at the proximal user (decoding private message) is given by:

$$\Upsilon_{K,p} = \frac{\left|(\mathbf{h}_k + \tilde{\mathbf{h}}_{vz}^k \Theta_{v,z} \tilde{\mathbf{H}}_{vz})\sqrt{\alpha_{K,p} p_K}\right|^2}{\left|(\mathbf{h}_k + \tilde{\mathbf{h}}_{vz}^k \Theta_{v,z} \tilde{\mathbf{H}}_{vz})\sqrt{\alpha_{L,p} p_L}\right|^2 + \sigma_n^2}, \quad (9)$$

$$R_{K,p}[\Upsilon_{K,p}] = T_{\tau_K} \log_2(1 + \Upsilon_{K,p}). \quad (10)$$

Similarly, the received SINR and the corresponding achievable rate at the distant user (decoding common message) is given by:

$$\Upsilon_{L,c} = \frac{\left|(\mathbf{h}_l + \tilde{\mathbf{h}}_{vz}^l \Theta_{v} \tilde{\mathbf{H}}_{vz})\sqrt{\alpha_{L,c} p_L}\right|^2}{\left|(\mathbf{h}_l + \tilde{\mathbf{h}}_{vz}^l \Theta_{v,z} \tilde{\mathbf{H}}_{vz})\sqrt{\alpha_{L,p} p_L}\right|^2 + \sigma_n^2}, \quad (11)$$

$$R_{L,c}[\Upsilon_{L,c}] = T_{\tau_L} \log_2(1 + \Upsilon_{L,c}), \quad (12)$$

while the received SINR and the corresponding achievable rate at distant user (decoding private message) is given by:

$$\Upsilon_{l,p} = \frac{\left|(\mathbf{h}_l + \tilde{\mathbf{h}}_{vz}^l \Theta_{v,z} \tilde{\mathbf{H}}_{vz})\sqrt{\alpha_{L,p} p_L}\right|^2}{\left|(\mathbf{h}_l + \tilde{\mathbf{h}}_{vz}^l \Theta_{v,z} \tilde{\mathbf{H}}_{vz})\sqrt{\alpha_{K,p} p_K}\right|^2 + \sigma_n^2}. \quad (13)$$

$$R_{L,p}[\Upsilon_{L,p}] = T_{\tau_L} \log_2(1 + \Upsilon_{L,p}). \quad (14)$$

The sum rate of distant user and proximal user for decoding common message is finally expressed as:

$$R_{u,sum} = \sum_{\Delta=K,L} T_{\tau_\Delta}(R_{\Delta,p} + R_{\Delta,p}). \quad (15)$$

D. Problem Formulation

In this paper, the joint optimization of power allocation and time allocation factors for the total channel capacity in a R-RS communication scenario is investigated, aiming to maximize the overall rate while ensuring power budget constraints for both proximal and distant users from the BS. In the first phase, a time-slot-based optimization method for the R-RS system channel time allocation factor is proposed in III-A. In the second phase, to ensure the maximization of channel capacity in the R-RS system while implementing an effective resource management strategy, a joint optimization method for the transmission power of training symbols p_t and data symbols p_d is introduced. This approach enables the system to flexibly allocate power between the training and data transmission phases, thereby saving power and reducing energy consumption. The mathematical formulation for describing this problem scenario is expressed as follows:

$$\begin{aligned}
&\max_{\{p_t, p_d, T_{\tau_L}, T_{\tau_K}\}} \sum R_{u,sum}\left(p_t, p_d, T_{\tau_L}, T_{\tau_K}\right), \\
&\text{s.t. C1}: P = p_t + p_d, \\
&\quad \text{C2}: \mathbf{s} = [\mathbf{s}_K, \mathbf{s}_L]^T, \mathbb{E}\left[\mathbf{s}\mathbf{s}^H\right] = \mathbf{I}, \\
&\quad \text{C3}: \Upsilon_K\left(p_K\right) = \textstyle\sum_{*=p,c} \Upsilon_{K,*}\left(p_K\right), \\
&\qquad \Upsilon_L\left(p_L\right) = \textstyle\sum_{*=p,c} \Upsilon_{L,*}\left(p_L\right), \\
&\quad \text{C4}: T_{\tau_K} = \chi T_{\tau_{total}}, \quad T_{\tau_L} = (1-\chi) T_{\tau_{total}}, \\
&\qquad \chi \in [0, 1], \\
&\quad \text{C5}: \left|\phi\left(\theta_{vz}\right)\right| = 1, \forall z \in [1, Z],
\end{aligned} \quad (16)$$

where P represents the total transmit power at the BS, p_t and p_d denote the transmit power for training symbol and data symbol need to be optimized in III-B. In the context of C3, $(\Upsilon_{\min})_K$ and $(\Upsilon_{\min})_L$ represent the SNR threshold values for proximal and distant users, below which communication transmission is not feasible. Then the achievable rate of proximal and distant users are expressed as $\log_2(1 + \Upsilon_K)$ and $\log_2(1 + \Upsilon_L)$, and $\log_2(1 + \Upsilon_K) \geq \log_2[1 + (\Upsilon_{\min})_K]$ and $\log_2(1 + \Upsilon_L) \geq [(\Upsilon_{\min})_L]$ need to be satisfied. P and $T_{\tau_{total}}$ in C1 and C4 are fixed values. In C4, T_{τ_K} and T_{τ_L} represent the transmission time matrices for proximal and distant users, $T_{\tau_{total}}$ indicates the total transmission time matrix, where χ signifies the allocation coefficient for the transmission time of the two users. C5 ensures that the amplitude of the reflection coefficient of each RIS element is 1.

III. Algorithm of RISs Assisted RSMA Rate Optimization Method

This section mainly introduces the proposed algorithm of STMMSE combined with RS-OPC (SR-O) method to optimize the channel capacity of R-RS system, as shown in Fig. 2 and Algorithm 1, where in the first stage of BS-RIS transmission, the signal transmission is divided according to the power distribution coefficients $\alpha_{K,p}$, $\alpha_{K,c}$, $\alpha_{L,p}$ and $\alpha_{L,c}$ of distant and proximal users and public and private streams. In the second stage of RIS-UE transmission, power allocation is carried out for users according to the power allocation coefficient $\{\alpha_1, \ldots, \alpha_U\}$ obtained by reflection link power allocation optimization method in III-C. In the WR-OPC scheme proposed in this paper, the time allocation of remote users and local users is optimized at the BS end, and then the power allocation is optimized according to the training symbols and data symbols. The optimized channel capacity algorithm proposed in this section is based on the RIS-assisted RSMA system as shown in Fig. 3. The transmission symbol types are divided into training symbols and data symbols, and the time interval is divided into remote user transmission time and local user transmission time. Since RSMA is based on NOMA with rate segmentation technology,

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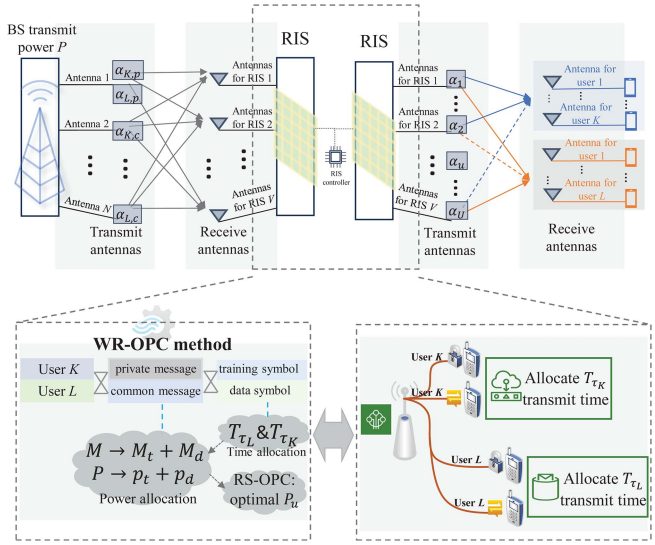

Fig. 2. Algorithm network of STMMSE and RS-OPC method assisted R-RS system.

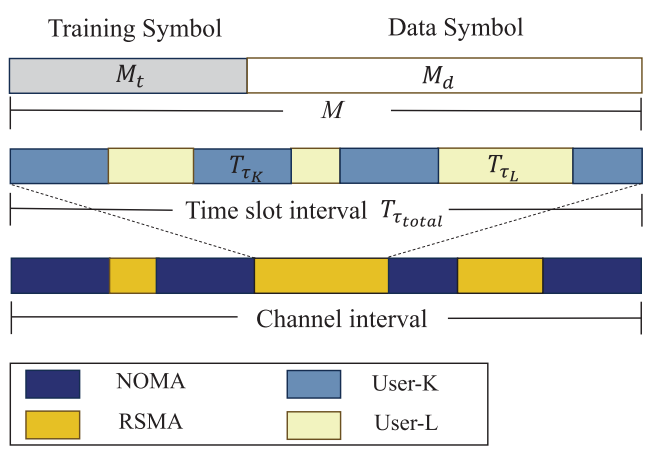

Fig. 3. Transmit symbols: including training symbols and data symbols. Time slot interval: including transmit time for proximal and distant users. Channel interval: including NOMA and RSMA. For the proposed method in this paper, the overall algorithm is built upon the RSMA communication system.

the channel interval is divided into NOMA and RSMA, and the transmission mode used in this paper is RSMA.

A. Time Allocation Coefficient Optimization

Firstly, the autocorrelation matrix as of the signal received by distant user is expressed:

$$\begin{aligned}\mathrm{R}_{\mathbf{y}_L} &= \mathbb{E}\left[\mathbf{y}_L\mathbf{y}_L^H\right]\\ &= \mathbb{E}\left\{\left[\left(\mathbf{h}_l+\tilde{\mathbf{h}}_{vz}^l\Theta_{v,z}\tilde{\mathbf{H}}_{vz}\right)\mathbf{s}_L+n_L\right]\right.\\ &\quad\left.\times\left[\left(\mathbf{h}_l+\tilde{\mathbf{h}}_{vz}^l\Theta_{v,z}\tilde{\mathbf{H}}_{vz}\right)\mathbf{s}_L+n_L\right]^H\right\}\\ &= \left(\mathbf{h}_l+\tilde{\mathbf{h}}_{vz}^l\Theta_{v,z}\tilde{\mathbf{H}}_{vz}\right)\mathbb{E}\left[\mathbf{s}_L\mathbf{s}_L^H\right]\\ &\quad\times\left(\mathbf{h}_l+\tilde{\mathbf{h}}_{vz}^l\Theta_{v,z}\tilde{\mathbf{H}}_{vz}\right)^H+\mathbb{E}\left[n_Ln_L^H\right]\\ &= \mathbf{H}_{RIS,L}\mathrm{R}_{\mathbf{s}_L}\mathbf{H}_{RIS,L}^H+\mathrm{R}_{n_L},\end{aligned}\tag{17}$$

where $\mathrm{R}_{\mathbf{s}_L}=\mathbb{E}[\mathbf{s}_L\mathbf{s}_L^H]$, $\mathrm{R}_{n_L}=\mathbb{E}[n_Ln_L^H]$. For brevity, $\mathbf{h}_l+\tilde{\mathbf{h}}_{vz}^l\Theta_{v,z}\tilde{\mathbf{H}}_{vz}$ is represented as $\mathbf{H}_{RIS,L}$, $\mathbf{h}_k+\tilde{\mathbf{h}}_{vz}^k\Theta_{v,z}\tilde{\mathbf{H}}_{vz}$ is represented as $\mathbf{H}_{RIS,K}$. Define the estimated signal with an STMMSE estimation method as $\hat{\mathbf{s}}_L=T_{\tau_L}\mathbf{y}_L$, which T_{τ_L} represents the time allocation coefficient matrix of distant user. The corresponding estimated error can be expressed as:

$$e_L=\hat{\mathbf{s}}_L-\mathbf{s}_L=T_{\tau_L}\mathbf{y}_L-\mathbf{s}_L.\tag{18}$$

According to the orthogonality principle where the error and the estimated value are orthogonal, which can be expressed as $\mathrm{R}_{\mathbf{s}_Ln_L}=\mathbb{E}[\mathbf{s}_Ln_L^H]=0$, $\mathrm{R}_{n_L\mathbf{s}_L}=\mathbb{E}[n_L\mathbf{s}_L^H]=0$, leading to a mean square error (MSE) as:

$$\begin{aligned}e_M &= \mathbb{E}\left\|T_{\tau_L}\mathbf{y}_L-\mathbf{s}_L\right\|^2\\ &= \mathbb{E}\left\{\mathrm{Tr}\left[\left(T_{\tau_L}\mathbf{y}_L-\mathbf{s}_L\right)\left(T_{\tau_L}\mathbf{y}_L-\mathbf{s}_L\right)^H\right]\right\}\\ &= \mathrm{Tr}\left\{T_{\tau_L}\mathbb{E}\left[\mathbf{y}_L\mathbf{y}_L^H\right]T_{\tau_L}^H-T_{\tau_L}\mathbb{E}\left[\mathbf{y}_L\mathbf{s}_L^H\right]\right.\\ &\quad\left.-\mathbb{E}\left[\mathbf{s}_L\mathbf{y}_L^H\right]T_{\tau_L}^H+\mathbb{E}\left[\mathbf{s}_L\mathbf{s}_L^H\right]\right\},\end{aligned}\tag{19}$$

where

$$\begin{cases}T_{\tau_L}\mathbb{E}\left[\mathbf{y}_L\mathbf{s}_L^H\right]=T_{\tau_L}\mathbb{E}[(\mathbf{H}_{RIS,L}\cdot\mathbf{s}_L+n_L)\mathbf{s}_L^H]\\ \qquad\qquad=T_{\tau_L}\mathbf{H}_{RIS,L}\mathrm{R}_{\mathbf{s}_L}\\ \mathbb{E}\left[\mathbf{s}_L\mathbf{y}_L^H\right]T_{\tau_L}^H=\mathbb{E}[\mathbf{s}_L(\mathbf{H}_{RIS,L}\cdot\mathbf{s}_L+n_L)^H]T_{\tau_L}^H\\ \qquad\qquad=\mathrm{R}_{\mathbf{s}_L}\mathbf{H}_{RIS,L}^HT_{\tau_L}^H.\end{cases}\tag{20}$$

Therefore, (19) can be reformulated as:

$$\begin{aligned}re_M = \mathrm{Tr}\left\{T_{\tau_L}\mathrm{R}_{\mathbf{y}_L}T_{\tau_L}^H-T_{\tau_L}\mathbf{H}_{RIS,L}\mathrm{R}_{\mathbf{s}_L}\right.\\ \left.-\mathrm{R}_{\mathbf{s}_L}\mathbf{H}_{RIS,L}^HT_{\tau_L}^H+\mathrm{R}_{\mathbf{s}_L}\right\}.\end{aligned}\tag{21}$$

By the properties of conjugate matrices, i.e., $(A^H)^T=A^*$, the following equation can be derived:

$$\begin{cases}\dfrac{\partial\mathrm{Tr}\left\{T_{\tau_L}\mathrm{R}_{\mathbf{y}_L}T_{\tau_L}^H\right\}}{\partial T_{\tau_L}}=\left(\mathrm{R}_{\mathbf{y}_L}T_{\tau_L}^H\right)^T=T_{\tau_L}^*\mathrm{R}_{\mathbf{y}_L}^T,\\ \dfrac{\partial\mathrm{Tr}\left\{T_{\tau_L}\mathbf{H}_{RIS,L}\mathrm{R}_{\mathbf{s}_L}\right\}}{\partial T_{\tau_L}}=\left(\mathbf{H}_{RIS,L}\mathrm{R}_{\mathbf{s}_L}\right)^T,\\ \dfrac{\partial\mathrm{Tr}\left\{\mathrm{R}_{\mathbf{s}_L}\mathbf{H}_{RIS,L}^HT_{\tau_L}^H\right\}}{\partial T_L}=0,\\ \dfrac{\partial\mathrm{Tr}\left\{\mathrm{R}_{\mathbf{s}_L}\right\}}{\partial T_{\tau_L}}=0.\end{cases}\tag{22}$$

The following results can be obtained by setting $\dfrac{\partial e_M}{\partial T_{\tau_L}}=0$:

$$T_{\tau_L}^*\mathrm{R}_{\mathbf{y}_L}^T=\left(\mathbf{H}_{RIS,L}\mathrm{R}_{\mathbf{s}_L}\right)^T.\tag{23}$$

Taking the complex conjugation on both sides of the equality sign yields:

$$T_{\tau_L}\mathrm{R}_{\mathbf{y}_L}^H=\mathrm{R}_{s_L}^H\mathbf{H}_{RIS,L}^H.\tag{24}$$

By the property of a Hermitian matrix, i.e., $\mathrm{R}_{\mathbf{s}_L}^H=\mathrm{R}_{\mathbf{s}_L}$ and $\mathrm{R}_{\mathbf{y}_L}^H=\mathrm{R}_{\mathbf{y}_L}$, (24) can be expressed as:

$$T_{\tau_L}\mathrm{R}_{\mathbf{y}_L}=\mathrm{R}_{\mathbf{s}_L}\mathbf{H}_{RIS,L}^H.\tag{25}$$

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Finally, the time allocation coefficient matrix of distant and proximal user are expressed as:

$$T_{\tau_L} = \mathrm{R}_{\mathbf{s}_L}\mathbf{H}_{RIS,L}^H\mathrm{R}_{\mathbf{y}_L}^{-1} = \mathrm{R}_{\mathbf{s}_L}\mathbf{H}_{RIS,L}^H\left(\mathbf{H}_{RIS,L}\mathrm{R}_{\mathbf{s}_L}\mathbf{H}_{RIS,L}^H + \mathrm{R}_{n_L}\right)^{-1}, \tag{26}$$

$$T_{\tau_K} = \mathrm{R}_{\mathbf{s}_K}\mathbf{H}_{RIS,K}^H\mathrm{R}_{\mathbf{y}_K}^{-1} = \mathrm{R}_{\mathbf{s}_K}\mathbf{H}_{RIS,K}^H\left(\mathbf{H}_{RIS,K}\mathrm{R}_{\mathbf{s}_K}\mathbf{H}_{RIS,K}^H + \mathrm{R}_{n_K}\right)^{-1}. \tag{27}$$

Assume that in this process, the estimated channel of the distant user is $\hat{\mathbf{H}}_{RIS,L} = \mathbf{H}_{RIS,L} + \mathbf{E}_{RIS,L}$, where $\mathbf{E}_{RIS,L}$ is the channel estimation error. The impact of $\mathbf{E}_{RIS,L}$ on the time allocation coefficient matrix T_{τ_L} is expressed as:

$$\hat{T}_{\tau_L} = \left(\hat{\mathbf{H}}_{L,RIS}^H\hat{\mathbf{H}}_{L,RRS} + \sigma_n^2\mathbf{I}\right)^{-1}\left(\hat{\mathbf{H}}_{L,RIS} + \mathbf{E}_{L,RIS}\right)^H. \tag{28}$$

When the channel is an unbiased estimator, the time allocation coefficient is also an unbiased estimator.

B. Power Allocation Optimization Method

Without loss of generality, a general data structure consisting of $M_{u,t}$ training symbols and $M_{u,d}$ data symbols for each spatial stream is considered. In the considered problem, the total length M_u of the transmitted symbols is predefined. The primary objective in this section is to allocate different transmission powers to training symbols and data symbols. Then, the RS-OPC method as shown in Algorithm 2 in detail, is used to derive the power distribution of the proximal and distant users in the data symbol transmission stage to maximize the total channel capacity. Define a weighting value ω, $0 < \omega < 1$, and have $p_t = \omega p = NM_{u,t}p_{t,ave}^2$ and $p_d = (1-\omega)p = NM_{u,d}p_{d,\mathrm{ave}}^2$.

The matrix N_d' in (4) combines channel estimation errors and noise, regarded as an equivalent additive noise, $N_d' \sim \mathcal{CN}(0, \sigma_{n'}^2)$, as shown in (29) and (30)~(34), shown at the bottom of the page, where $p_{\Delta,ave}^2$ denote the average transmit power of channel estimated error, i.e., $\mathbb{E}[\Delta\mathbf{H}_{RIS}\Delta\mathbf{H}_{RIS}^H] = Np_{\Delta,ave}^2\mathbf{I}_N$.

$$\mathbb{E}\left[N_d'N_d'^H\right] = \mathbb{E}\left[\Delta\mathbf{H}_{RIS}\mathbf{S}_d\mathbf{S}_d^H\Delta\mathbf{H}_{RIS}^H\right] + \mathbb{E}\left[N_dN_d^H\right], \tag{29}$$

With an MMSE multi-channel estimation, the estimated channel matrix during the training symbol process is:

$$\begin{aligned}\hat{\mathbf{H}}_{RIS} &= \mathbf{H}_{RIS} - \Delta\mathbf{H}_{RIS}\\ &= \mathbf{H}_{RIS}\mathbf{S}_t\mathbf{S}_t^H\left(p_{\Delta,ave}^2\mathbf{I}_N + \mathbf{S}_t\mathbf{S}_t^H\right)^{-1}\\ &\quad + N_tN_t^H\left(p_{\Delta,ave}^2\mathbf{I}_N + \mathbf{S}_t\mathbf{S}_t^H\right)^{-1},\end{aligned} \tag{35}$$

where $\mathbf{S}_t \in \mathbb{C}^{N\times M_{u,t}}$ is the training symbol matrix with the average transmit power $p_{t,ave}^2$. Define $V_H = \frac{1}{N}\mathbb{E}[\mathbf{H}_{RIS}^H\mathbf{H}_{RIS}] \in \mathbb{C}^{1\times N}$, take the singular value decomposition (SVD) of V_H, $V_H = U_H\Sigma_HU_H^H$, where $\Sigma_H = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_N)$, and $\frac{1}{N}\sum_{i=1}^N\lambda_i = \frac{1}{N}\mathrm{Tr}(V_H) = p_{h,ave}^2$, $i = 1, \ldots, N$. From (34), $\mathbb{E}[\Delta\mathbf{H}_{RIS}\Delta\mathbf{H}_{RIS}^H]$ can be derived as:

$$\begin{aligned}&\mathbb{E}[\Delta\mathbf{H}_{RIS}\Delta\mathbf{H}_{RIS}^H] = \mathbb{E}[(\mathbf{H}_{RIS} - \hat{\mathbf{H}}_{RIS})(\mathbf{H}_{RIS} - \hat{\mathbf{H}}_{RIS})^H]\\ &\geq \Sigma_{CRLB} = \left(\frac{\mathbf{S}_t\mathbf{S}_t^H}{\sigma_n^2} + V_{\mathbf{H}}^{-1}\right)^{-1} = \left(\frac{U_{\mathbf{H}}^H\mathbf{S}_t\mathbf{S}_t^HU_{\mathbf{H}}}{\sigma_n^2} + \Sigma_{\mathbf{H}}^{-1}\right)^{-1}\\ &= \mathrm{diag}\left(\frac{\sigma_n^2\lambda_1}{\sigma_n^2 + M_{u,t}p_{t,ave}^2\lambda_1}, \ldots, \frac{\sigma_n^2\lambda_i}{\sigma_n^2 + M_{u,t}p_{t,ave}^2\lambda_t},\right.\\ &\quad \left.\ldots, \frac{\sigma_n^2\lambda_N}{\sigma_n^2 + M_{u,t}p_{t,ave}^2\lambda_N}\right),\end{aligned} \tag{36}$$

which is the Cramér-Rao lower bound (CRLB) of the channel matrix estimation. Based on (36), a lower bound of $\mathrm{R}_{\Delta\mathbf{H}_{RIS}}$ is given by $\mathcal{M}_L = \mathrm{Tr}(\Sigma_{CRLB}) = \sum_{i=1}^N\frac{\sigma_n^2\lambda_i}{\sigma_n^2 + M_{u,t}p_{t,ave}^2\lambda_i}$. $\mathcal{M}_L$ can be seen as a function of λ_i with the constraint of $\sum_{i=1}^N\lambda_i = \mathrm{Tr}(V_{\mathbf{H}}) = N\cdot p_{h,ave}^2$. Considering the average channel gain, it is necessary to use the Lagrange multiplier method to find the lower bound of $\mathcal{M}_L$ in order to minimize $\mathcal{M}_L$. The Lagrange function is given by:

$$\mathcal{L}(\Sigma_{\mathbf{H}}) = \sum_{i=1}^N\frac{\sigma_n^2\lambda_i}{\sigma_n^2 + M_{u,t}p_{t,ave}^2\lambda_i} + \alpha\left[\sum_{i=1}^N\lambda_i - \mathrm{Tr}(V_{\mathbf{H}})\right], \tag{37}$$

where α is the Lagrange multiplier. By solving $\frac{\partial\mathcal{L}(\Sigma_{\mathbf{H}})}{\partial\lambda_i} = 0$, get $\frac{\sigma_n^4}{(\sigma_n^2 + M_{u,t}p_{t,ave}^2\lambda_i)^2} + \alpha = 0$, which indicates that the lower bound is achieved when $\lambda_i = \frac{1}{N}\mathrm{Tr}(V_{\mathbf{H}}) = \frac{1}{N}\sum_{i=1}^N\lambda_i =$

$$\mathbb{E}\left[N_d'N_d'^H\right] = \mathbb{E}\left\{\begin{array}{l}\left[\left(\sum_{\star=k,l}(\mathbf{h}_{1,\star} + \Delta\tilde{\mathbf{h}}_{1,vz}^{\star}\Theta_{v,z}\Delta\tilde{\mathbf{H}}_{1,vz})\right), \ldots, \left(\sum_{\star=k,l}(\mathbf{h}_{j,*} + \Delta\tilde{\mathbf{h}}_{j,vz}^{\star}\Theta_{v,z}\Delta\tilde{\mathbf{H}}_{j,vz})\right), \ldots,\\ \left(\sum_{\star=k,l}(\mathbf{h}_{N,\star} + \Delta\tilde{\mathbf{h}}_{N,vz}^{\star}\Theta_{v,z}\Delta\tilde{\mathbf{H}}_{N,vz})\right)\right]^T\mathbf{S}_d\mathbf{S}_d^H\left[\left(\sum_{\star=k,l}(\mathbf{h}_{1,*} + \Delta\tilde{\mathbf{h}}_{1,vz}^{\star}\Theta_{v,z}\Delta\tilde{\mathbf{H}}_{1,vz})\right)^*, \ldots,\\ \left(\sum_{\star=k,l}(\mathbf{h}_{j,\star} + \Delta\tilde{\mathbf{h}}_{j,vz}^{\star}\Theta_{v,z}\Delta\tilde{\mathbf{H}}_{j,vz})\right)^*, \ldots, \left(\sum_{\star=k,l}(\mathbf{h}_{N,\star} + \Delta\tilde{\mathbf{h}}_{N,vz}^{\star}\Theta_{v,z}\Delta\tilde{\mathbf{H}}_{N,vz})\right)^*\right]\end{array}\right\} + \mathbb{E}\left[N_dN_d^H\right] \tag{30}$$

$$\mathbb{E}[N_d'N_d'^H] = \mathbb{E}\left\{[\Delta\mathbf{H}_{1,RIS}, \ldots, \Delta\mathbf{H}_{j,RIS}, \ldots, \Delta\mathbf{H}_{N,RIS}]^T\mathbf{S}_d\mathbf{S}_d^H[\Delta\mathbf{H}_{1,RIS}^*, \ldots, \Delta\mathbf{H}_{j,RIS}^*, \ldots, \Delta\mathbf{H}_{N,RIS}^*]\right\} + \mathbb{E}[N_dN_d^H] \tag{31}$$

$$\mathbb{E}[N_d'N_d'^H] = \mathrm{diag}\left\{\mathrm{Tr}\left(\mathbf{S}_d\mathbf{S}_d^H\mathbb{E}\left[\Delta\mathbf{H}_{1,RIS}^*\Delta\mathbf{H}_{1,RIS}^T\right]\right), \ldots, \mathrm{Tr}\left(\mathbf{S}_d\mathbf{S}_d^H\mathbb{E}\left[\Delta\mathbf{H}_{j,RIS}^*\Delta\mathbf{H}_{j,RIS}^T\right]\right)\right. \tag{32}$$

$$\left.\ldots, \mathrm{Tr}\left(\mathbf{S}_d\mathbf{S}_d^H\mathbb{E}\left[\Delta\mathbf{H}_{N,RIS}^*\Delta\mathbf{H}_{N,RIS}^T\right]\right)\right\} + M_d\sigma_n^2\mathbf{I}_N \tag{33}$$

$$\mathbb{E}[N_d'N_d'^H] = NM_dp_{d,ave}^2p_{\Delta,ave}^2\mathbf{I}_N + M_d\sigma_n^2\mathbf{I}_N \triangleq M_d\sigma_{n'^2}\mathbf{I}_N \tag{34}$$

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$p_{h,cve}^2, i = 1, \ldots, N$. The lower bound of $\mathcal{M}_L$ is:

$$\mathcal{M}_L \geq \frac{N \cdot \sigma_n^2 p_{h,ave}^2}{\sigma_n^2 + M_{u,t} p_{t,ave}^2 p_{h,ave}^2} = \mathcal{M}_{L,\min}. \tag{38}$$

For any diagonal element of $\Sigma_{CRLB}(i)$, we can get:

$$\Sigma_{CRLB}(i) \geq \frac{\sigma_n^2 p_{h,ave}^2}{\sigma_n^2 + M_{u,t} p_{t,ave}^2 p_{h,ave}^2} = M_e, \quad i = 1, 2, \ldots, N. \tag{39}$$

The condition for the equality sign is when V_H is a scaled identity matrix, i.e., $\lambda_1 = \cdots = \lambda_N$. Using the lower bound of the obtained minimum value, subsequent power allocations can be evaluated to maximize channel capacity.

Theorem 1: When the lower bound of $\mathcal{M}_L$ is achieved, the upper bound of SNR is expressed as:

$$\Upsilon_{up} = \begin{cases} \dfrac{\Upsilon M_u}{M_{u,d} - N}(\sqrt{\Lambda} - \sqrt{\Lambda - 1}), & M_{u,d} > N, \\ \dfrac{(\Upsilon M_u)^2}{4N(N + \Upsilon M_u)}, & M_{u,d} = N, \\ \dfrac{\Upsilon M_u}{N - M_{u,d}}(\sqrt{-\Lambda} - \sqrt{-\Lambda + 1}), & M_{u,d} < N. \end{cases} \tag{40}$$

The proof is provided in Appendix A.

At high and low SNR, the parameters and optimal power allocation is:

$$\Lambda_h = \frac{M_{u,d}}{M_{u,d} - N}, \qquad \Lambda_l = \frac{N M_{u,d}}{\Upsilon M_u (M_{u,d} - N)}, \tag{41}$$

$$\omega_{op,h} = \frac{\sqrt{M_{u,d}}}{\sqrt{M_{u,d}} + \sqrt{N}}, \qquad \omega_{op,l} = \frac{1}{2}. \tag{42}$$

The corresponding SNR will be reformulated as:

$$\Upsilon_{up,h} = \frac{M_u}{(\sqrt{M_{u,d}} + \sqrt{N})^2}\Upsilon, \quad \Upsilon_{up,l} = \frac{M_u^2}{4 M_u M_{u,d}}\Upsilon^2. \tag{43}$$

Since $\hat{\mathbf{H}}_{RIS} = \mathbf{H}_{RIS} - \Delta\mathbf{H}_{RIS}$, $\hat{\mathbf{H}}_{RIS}$ is a random variable with zero mean and variance $\sigma_{\hat{\mathbf{H}}_{RIS}}^2 = \frac{1}{N^2}\mathbb{E}\{\mathrm{Tr}[\hat{\mathbf{H}}_{RIS}\hat{\mathbf{H}}_{RIS}^H]\}$. Based on MMSE principle, $\sigma_{\hat{\mathbf{H}}_{RIS}}^2 = p_{h,ave}^2 - p_{\Delta,ave}^2$ can be obtained, and $\hat{\mathbf{H}}_{RIS}$ can be normalized as $\tilde{\mathbf{H}}_{RIS} = \frac{1}{\sigma_{\hat{\mathbf{H}}_{RIS}}^2}\hat{\mathbf{H}}_{RIS}$. Due to the lower bound of $\mathcal{M}_L$ in (38), the total channel capacity $R_u(P)$ can be upper bound by:

$$R_{u,sum}(P) \leq \underbrace{\sum_{u=1}^{U} \log_2 \left[\frac{(p_{h,ave}^2 - M_e)\mathbf{H}_{RIS}}{\dfrac{p_d}{M_{u,d}} M_e + \sigma_n^2} + 1 \right]}_{R_{u,upper}(P)}. \tag{44}$$

Let $\frac{1}{N}\mathbb{E}[\tilde{\mathbf{H}}_{RIS}\tilde{\mathbf{H}}_{RIS}^H] = U_{\tilde{\mathbf{H}}}\Sigma_{\tilde{\mathbf{H}}}U_{\tilde{\mathbf{H}}}^H$, which $\Sigma_{\tilde{\mathbf{H}}} = \mathrm{diag}\{\beta_{11}, \ldots, \beta_{ii}, \ldots, \beta_{NN}\}$ is a diagonal matrix. Let $\Gamma = (\mathbf{S}_d U_{\tilde{\mathbf{H}}})^H \mathbf{S}_d U_{\tilde{\mathbf{H}}} = (Y_d)^H Y_d$ with (i,j)-th entry being $\mu_{i,j}$. According to Hadamard's inequality, i.e., $\det(\Gamma_{N\times N}) \leq \prod_{i=1}^{N} \mu_{ii}$. The maximum value of $R_{u,upper}(P)$ is derived as:

$$R_{u,\max}(P) = \max \sum_{u=1}^{U}\sum_{i=1}^{N} \left\{ \log_2 \left[\frac{(p_{h,ave}^2 - M_e)\beta_{ii}\mu_{ii}}{\dfrac{P_d}{M_{u,d}} M_e + \sigma_n^2} + 1 \right] \right\},$$
$$\text{s.t.} \quad \mathrm{Tr}(\Gamma) \leq \omega_{op} \cdot P,$$
$$\mu_{ii} \geq 0, \quad 1 \leq i \leq N. \tag{45}$$

Introducing α' associated with μ_{ii}, the optimal solution can be obtained based on Lagrange multiplier as:

$$\mu_{ii} = \left[-\frac{1}{\alpha' \ln 2} - \frac{\sigma_n^2 + \dfrac{P_d}{M_{u,d}} M_e}{\beta_{ii}(p_{h,ave}^2 - M_e)} \right]. \tag{46}$$

Algorithm 1: The Proposed SR-O Method in R-RS System.

Input: $\mathbf{H}_{RIS,K}$, $\mathbf{H}_{RIS,L}$, P, α'
for $\Delta = k, l$ **do**
  Optimize with respect to T_{τ_Δ} given P;
  for $v = 1, 2, \ldots, V$ **do**
    $\mathbf{H}_{RIS,\Delta} = \mathbf{h}_\Delta + \tilde{\mathbf{h}}_{vz}^\Delta \Theta_{v,z} \tilde{\mathbf{H}}_{vz}$;
    $\mathrm{R}_{y_L} = \mathbb{E}[y_L y_L^H]$;
    Compute R_{y_L} as in (17);
    According to $e_M = 0$;
    $\dfrac{\partial e_M}{\partial \tau_{\tau_K}} = 0, \dfrac{\partial e_M}{\partial \tau_{\tau_L}} = 0$;
    Output: T_{τ_K}, T_{τ_L}
  end
  Optimize with respect to P given T_{τ_Δ};
  Divide symbol into training and data;
  for *data* **do**
    $Y_d \leftarrow N_d'$;
    Compute $\mathrm{R}_{N_d'}$, get $p_{\Delta,ave}^2$;
  end
  for *training* **do**
    $\hat{\mathbf{H}}_{RIS} \leftarrow \mathrm{R}_{\Delta H_{RIS}}$;
    Compute lower bound M_e of Σ_{CRLB} as in (36);
  end
  Maximizing using M_e;
  $\mathcal{L}(R_u) = \max R_{u,upper}(P) + \alpha'[\mathrm{Tr}(\Gamma) - \omega_{op}P]$;
  Output: $R_{u,max}(P)$
end

C. RIS Phase Shift Optimization and Reflection Link Power Allocation

For the downlink reflection link in the RIS-UE system, a second-stage power allocation is required at the RIS side. The purpose of distinguishing between the BS-side power allocation and the RIS-side power allocation is to optimize the power distribution factors at both ends separately in order to overcome fading and interference, thereby improving the signal quality of

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Algorithm 2: The Proposed RS-OPC Method.

Input: $\tilde{L}_u$, P, α_u

This method targets all users in the system, including proximal and distant users, the set of users are denoted as;

$$\mathcal{U} = \{1, ..., u, ..., U\};$$
$$\hat{\mu} = \left(P + \sum_{u\in\mathcal{U}} \tilde{L}_u\right) / \left(\sum_{u\in\mathcal{U}} \alpha_u\right);$$
$$\hat{k} = \arg\max_{u\in U}(\tilde{L}_u/\alpha_u);$$

According to stationarity conditions and complementary slackness conditions, the comparison below can be obtained;

while $(\tilde{L}_{\hat{k}}/\alpha_{\hat{k}}) \geq \hat{\mu}$ **do**

$\quad \mathcal{U} = \mathcal{U} - \hat{k};$

$\quad (P_u/\alpha_u) = 0;$

end

while $(\tilde{L}_{\hat{k}}/\alpha_{\hat{k}}) < \hat{\mu}$ **do**

$\quad (P_u/\alpha_u) = \hat{\mu} - (\tilde{L}_u)/(\alpha_u)$

end

The power allocation result for user u is represented as;

Output: P_u

the system. However, before performing the power allocation at the RIS side, it is necessary to optimize the RIS phase shifts to achieve channel gain maximization. The optimized RIS phase shifts for the distant l-th user and the proximal k-th user are obtained as:

$$\left(\theta^*_{v,z}\right)_l = \arg\left(\mathbf{h}_l\right) - \arg\left(\left[\tilde{\mathbf{H}}_{vz}\right]_z \left[\tilde{\mathbf{h}}^l_{vz}\right]_z\right),$$
$$\left(\theta^*_{v,z}\right)_k = \arg\left(\mathbf{h}_k\right) - \arg\left(\left[\tilde{\mathbf{H}}_{vz}\right]_z \left[\tilde{\mathbf{h}}^k_{vz}\right]_z\right), \tag{47}$$

where $\arg(\mathbf{h}_l)$ and $\arg(\mathbf{h}_k)$ represent the phases of the signals transmitted over the direct links $\mathbf{h}_l$ and $\mathbf{h}_k$ between the proximal and distant users, $\arg([\tilde{\mathbf{H}}_{vz}]_z[\tilde{\mathbf{h}}^l_{vz}]_z)$ and $\arg([\tilde{\mathbf{H}}_{vz}]_z[\tilde{\mathbf{h}}^k_{vz}]_z)$ represent the phases of the signals transmitted over the cascaded links assisted by RIS for the proximal and distant users. The difference between the two is the phase offset $(\theta^*_{v,z})_l$ and $(\theta^*_{v,z})_k$ that needs to be compensated. The purpose is to eliminate the phase distortion during signal transmission or processing through this parameter, making the phase of the received signal consistent with that of the reference signal.

For the proof see Appendix B.

Define the total power P where the power allocation matrix for each user is $P = \mathrm{diag}\{P_1, \ldots, P_u\}$, $P_u = [P_c, P_p] = [\alpha_{u,c}P_u, \alpha_{u,p}P_u]$, $P_c = \mathrm{diag}\{P_{c,1}, \ldots, P_{c,u}\}$, $P_p = \mathrm{diag}\{P_{p,1}, \ldots, P_{p,u}\}$, $u = 1, \ldots, U$. By introducing the power allocation coefficient and Lagrange multiplier of user u represented by α_u and μ', the optimal power distribution scheme of distant user K and proximal user L is obtained by RS-OPC method as $P_K = \frac{\alpha_K}{\mu'} - \tilde{L}_K$, $P_L = \frac{\alpha_L}{\mu'} - \tilde{L}_L$. It is interpreted that the power for distant users to transmit both common and private message in the transmission data symbol stage is denoted as $P_{K,c} = P_{K,p} = P_K$, the power for proximal users to transmit both common and private message is expressed as $P_{L,c} = P_{L,p} = P_L$, see Appendix C for proof.

D. Analysis of Optimality, Convergence, and Computational Complexity

For the STMMSE and RS-OPC algorithms proposed in this paper, the optimality, convergence and computational complexity of the algorithms are analyzed in detail from the calculation operations of the main computational steps and their corresponding scaling relationships.

1) Optimality of STMMSE and RS-OPC: The STMMSE algorithm is designed to minimize the MSE in channel estimation while optimizing time and power allocation, while RS-OPC aims to maximize the total channel capacity by optimizing the power allocation between the common and private streams in RSMA as shown in Table I.

2) Convergence of STMMSE and RS-OPC: The convergence of STMMSE can be analyzed mathematically based on the iterative optimization process, while the convergence of RS-OPC is based on the properties of convex optimization and iterative update rules as shown in Table I.

3) Computational Complexity of STMMSE: In the channel estimation stage, the dimension of channel matrix $\mathbf{H}_{RIS}$ is $1 \times N$, the dimension of covariance matrix is $N \times N$, and the complexity of calculating covariance matrix is $O(N^2)$. Inverse the covariance matrix of $N \times N$with complexity of $O(N^3)$. The complexity of calculating the channel estimation result $\hat{\mathbf{H}}_{RIS}$ is $O(N^2)$. In the allocation optimization stage, there are U users and Z RIS reflection units, and the complexity of a single iteration of the allocation optimization problem is $O(U \cdot Z)$, assuming that a total of I_1 iterations are required and the total complexity is $O(I_1 \cdot U \cdot Z)$. Combining channel estimation and allocation optimization, the total computational complexity of STMMSE is $O(I_1 \cdot U \cdot Z + N^3)$.

4) Computational Complexity of RS-OPC: In the power allocation optimization stage, assuming there are U users and V RISs, each iteration needs to calculate the power matrix P, and the complexity is $O(U \cdot V)$. Calculating the contribution of power to the channel capacity in each iteration, the complexity of multiplying the matrix $1 \times N$ with $N \times 1$ is $O(N)$. Assume that a total of I_2 iterations are required, and the total complexity is $O(I_2 \cdot U \cdot V \cdot N)$. After each power update, the channel capacity needs to be calculated, which involves matrix logarithm calculation with a complexity of $O(N^3)$. Combined power allocation and channel capacity calculation, the total complexity of RS-OPC is $O(I_2 \cdot U \cdot V \cdot N + N^3)$.

IV. Simulation Results

In this section, an in-depth simulation analysis of the R-RS system with proposed algorithm in this paper is presented. The simulation scenarios are configured as a system with Rayleigh fading featuring a multi-antenna base station interacting with multiple users and a variable number of RISs. Notably, in Figs. 4 and 5, the simulation maintains a fixed configuration with 6 transmit and receive antennas and a total of 4 users. The values of p_t and p_d are determined by the ω_{op} of *Theorem 1*, assuming

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TABLE I
ANALYSIS OF ALGORITHM COMPLEXITY

Algorithm	Optimality Metric	Achieved Optimality	Convergence Property
STMMSE	Minimizes Mean Squared Error (MSE)	• Achieves CRLB and optimal channel estimation accuracy. • Balances time and power allocation for maximum channel capacity.	• **Convergence in Channel Estimation:** The optimization minimizes a strictly convex function (MSE), ensuring convergence to a unique global minimum. The convergence rate depends on the condition number of R_{y_L}. • **Convergence in Allocation:** Convergence is guaranteed under standard conditions for convex optimization, and the convergence rate is typically sublinear.
RS-OPC	Maximizes Total Channel Capacity	• Ensures global optimality for power allocation using convex optimization and KKT conditions.	• **Convergence in Allocation:** Guaranteed convergence to the global optimum due to concavity of the objective function. Satisfies KKT conditions, with sublinear or quadratic convergence.

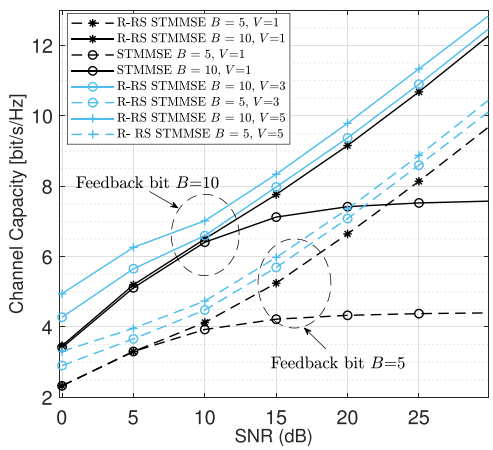

Fig. 4. Channel capacity versus SNR for different number of RIS in RIS-assisted RSMA system.

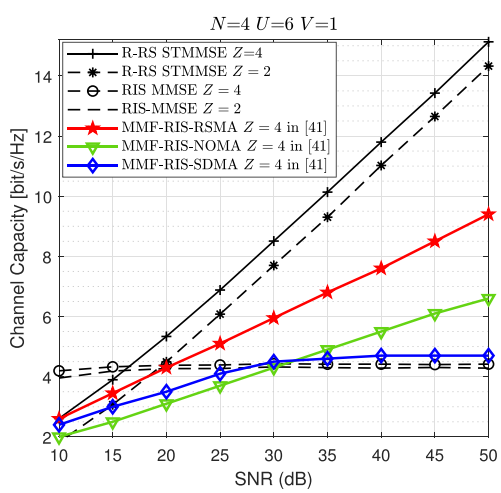

Fig. 5. Comparison of channel capacity versus SNR with other current advanced algorithms for RIS-RSMA scheme.

that the length of the training symbol is equal to the number of transmit antennas, i.e., $M_t = N$. The main parameters and values of simulations are summarized in Table II.

Fig. 4 illustrates the comparison of channel capacity trends with SNR for varying numbers of RIS using the STMMSE method optimizes channel capacity based on time-allocation coefficients. The SNR used in Figs. 4, 5 and 6 represents the average

TABLE II
MAIN PARAMETERS AND VALUES OF SIMULATION

Model Parameters	Value
Achievable rate ζ, [bits/s/Hz]	4
Amplitude reflection coefficient	1
The scale parameter of Rayleigh fading $\beta_{\mathbf{H}}$	1
The number of RIS reflection elements Z	10
The number of UE	4
The length of transmitted data M	5000
Bandwidth, [GHz]	10
Carrier frequency f_c, [GHz]	3
Gaussian white noise figure n_K and n_L, [dB]	10
Phase adjustment range	[0,2π]

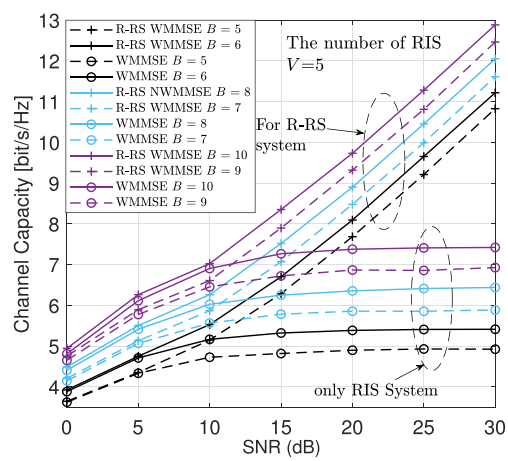

Fig. 6. Channel capacity comparison between RIS-assisted RSMA systems and RIS-assisted communication systems when the number of RIS is fixed at 5.

SNR of all users in the system and is expressed as $SNR_{\text{ave}} = 1/U[\sum_{k=1}^{K}(\Upsilon_{k,c} + \Upsilon_{k,p}) + \sum_{l=1}^{L}(\Upsilon_{l,c} + \Upsilon_{l,p})]$. As depicted in Fig. 4, an increase in the number of RISs results in a higher channel capacity for both feedback bit $B = 5$ and $B = 10$. Notably, the system incorporating RSMA technology exhibits significantly greater channel capacity gains when the SNR surpasses 10 dB, compared to scenarios without RSMA technology.

Fig. 5 describes the performance comparison between the STMMSE algorithm proposed in this paper and the max-min fair (MMF)-RIS-RSMA, MMF-RIS-SDMA and MMF-RIS-NOMA algorithms proposed in [45]. It can be seen from the figure that the STMMSE algorithm can obtain higher channel capacity compared with the algorithm of [45], which highlights the advantage of the STMMSE algorithm. Fig. 6 presents the

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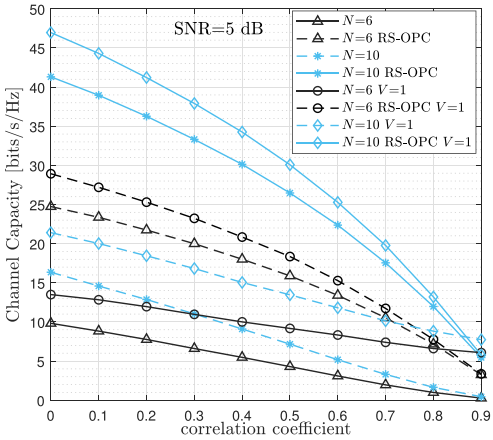

Fig. 7. Channel capacity comparison with RS-OPC algorithm: non-RIS vs. RIS-assisted RSMA Systems.

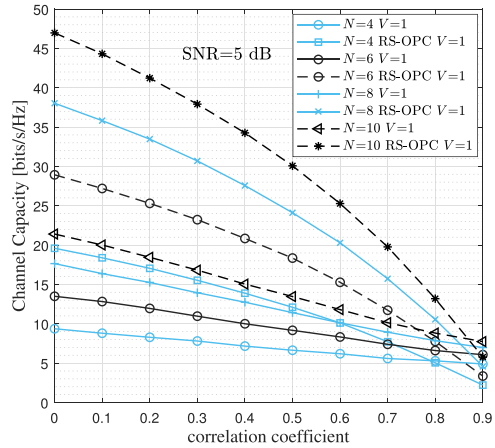

Fig. 8. Channel capacity of RSMA system when the number of RIS is fixed at 1.

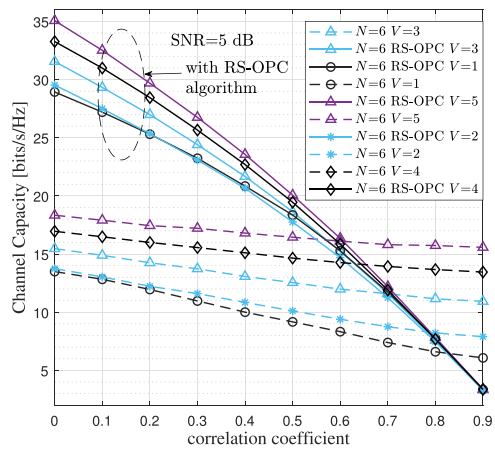

Fig. 9. Channel capacity with different number of RIS for the proposed RS-OPC algorithm.

outcomes of the comparison between the optimized channel capacity using the STMMSE algorithm in both RSMA and non-RSMA systems. The figure illustrates that, with a constant number of RIS, the channel capacity in the RSMA system surpasses that of without RSMA assistance when the SNR exceeds 10 dB. Moreover, as the feedback bit parameter B increases, the channel capacity within the RSMA system demonstrates a corresponding increase. That is because RSMA technology is able to overcome residual interference in multi-channels, solving the problem that traditional precoding cannot eliminate these interference.

Fig. 7 compares the channel capacity trends of RSMA systems with and without RIS assistance versus correlation coefficient under the RS-OPC optimal allocation method proposed in this paper, where correlation coefficient refers to the correlation between transmit and receive antennas. In the simulations conducted in this section, the correlation coefficient is set to range from 0 to 0.9. For varying numbers of transmit antennas in Fig. 7, the RSMA channel capacity with RIS assistance surpasses that without RIS assistance. Additionally, the channel capacity achieved through the optimized power allocation by the RS-OPC algorithm is higher than that obtained through average power allocation. As illustrated in Figs. 7, 8 and 9, the larger the correlation coefficient between antennas, the greater the loss in channel capacity. Fig. 8 illustrates a comparison of the variation in RIS-assisted RSMA channel capacity with and without RS-OPC algorithm-optimized power allocation, while keeping the number of RISs fixed at one. The figure highlights that the system's channel capacity, optimized through the RS-OPC power allocation algorithm, experiences a substantial increase with varying numbers of antennas. This enhancement is attributed to the RS-OPC algorithm's capability to dynamically allocate transmit power based on channel conditions, thereby maximizing the transmission rate.

Fig. 9 contrasts the channel capacity trends of the RSMA system assisted with varying numbers of RISs, while maintaining a constant number of transmit antennas. The comparison is made between the RS-OPC power allocation method and average power allocation. As the channel correlation coefficient increases incrementally, there is a gradual decrease in channel capacity. The figure clearly illustrates that, for different numbers of RISs, the optimized channel capacity achieved by the RS-OPC algorithm exhibits a significant advantage over the average power allocation method.

Fig. 10 examines the comparison of channel capacity after optimization through the RS-OPC method for RSMA systems with and without RIS assistance. When configured with varying numbers of transmit antennas, the RSMA system assisted by RIS achieves a higher channel capacity with RS-OPC algorithm. Importantly, the channel capacity of the system supported by the RS-OPC algorithm consistently outperforms that of the system without RS-OPC algorithm support across different configurations of transmit antennas. With a fixed number of RIS, Fig. 11 provides a comparison of channel capacity between two scenarios: optimization using the RS-OPC power allocation algorithm and average power allocation, considering various numbers of transmit antennas. The figure clearly illustrates that as the number of transmit antennas increases, the channel capacity of

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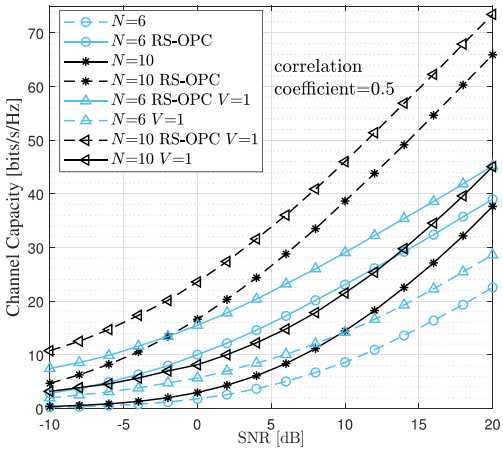

Fig. 10. Channel capacity versus SNR for the system with and without RIS.

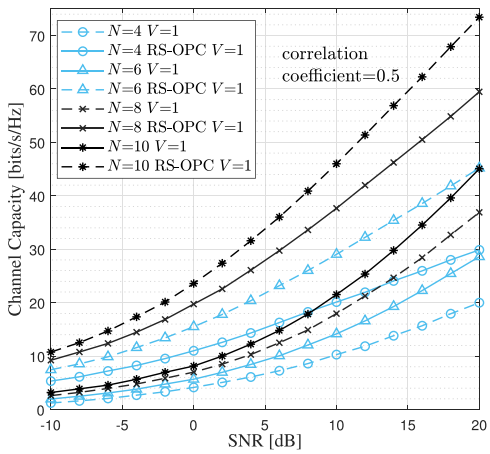

Fig. 11. Channel capacity versus SNR for different numbers of transmit antennas.

the RSMA system becomes greater, particularly when optimized with the RS-OPC algorithm.

The comparison in Fig. 12 depicts the trend of channel capacity changes achieved through RS-OPC and average power allocation methods, with a fixed number of transmitting antennas and varying numbers of RISs. The channel capacity variation in the RSMA system, ranging from 1 to 5 RIS numbers, highlights that the channel capacity consistently remains higher under the optimization of the RS-OPC power allocation algorithm when compared to average power allocation.

With the emergence of the IoT, there is an increasing focus on improving energy efficiency while striving for maximum network capacity. This entails minimizing the energy consumption of IoT devices and networks, mitigating the environmental footprint of IoT technologies, and optimizing resource allocation. The total power consumption for proposed algorithm in this paper, P_{total}, including both the transmit power P of BS and dissipation in hardware components of BSs, users and RISs, can be expressed as $P_{total} = \frac{P}{\eta} + P_e + P_u + NP_{BS}$, which η indicates the efficiency of the transmit power, P_e expresses the

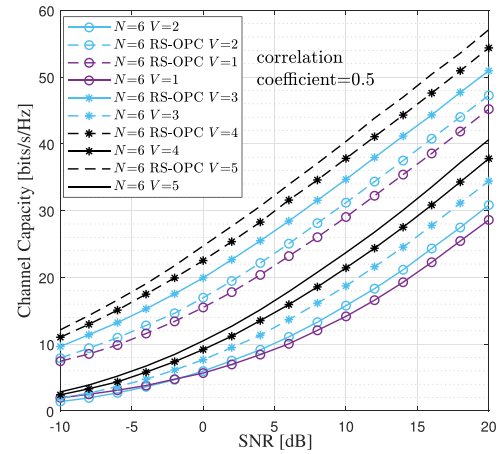

Fig. 12. Channel capacity versus SNR for different numbers of RISs.

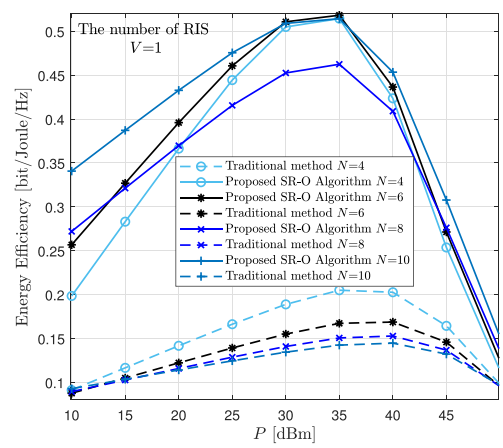

Fig. 13. Energy efficiency of the proposed SR-O method.

total power consumption caused by the circuitry required for adaptive phase shifting in RIS, P_u and P_{BS} are the hardware dissipation power at the BSs and users, respectively. Hence, the energy efficiency (EE) of the system can be described as $\eta_{EE} = BW\frac{\eta_{SE}}{P_{total}}$, which η_{SE} is the SE of the system expressed as $\eta_{SE} = N\sum_{u=1}^{U}\log(1+\Upsilon_u)$, BW is the bandwidth of the system and Υ_u denote the SNR for the u-th user.

Building upon the analysis mentioned above, Figs. 13 and 14 present simulation analyses of the proposed algorithm in terms of EE and SE versus transmit power P. Simulation parameters are set as $\eta = 0.38$, $P_{BS} = 30$ [dBm], $P_u = 40$ [dBm], $P_e = 40$ [dBm] and $BW = 10$ [MHz], and there is one RIS fixed in R-RS system. It can be observed from the curves that, for various numbers of transmit antennas, the algorithm presented in this paper consistently achieves high performance gains in the R-RS system. Fig. 15 compares the SE of the proposed algorithm in this paper with the algorithm and the continuous scheme presented in [50]. The comparison in the figure demonstrates the advantages of the proposed algorithm.

V. Conclusion

This paper has introduced a method for optimizing channel capacity in RIS-assisted MU-MISO communication systems

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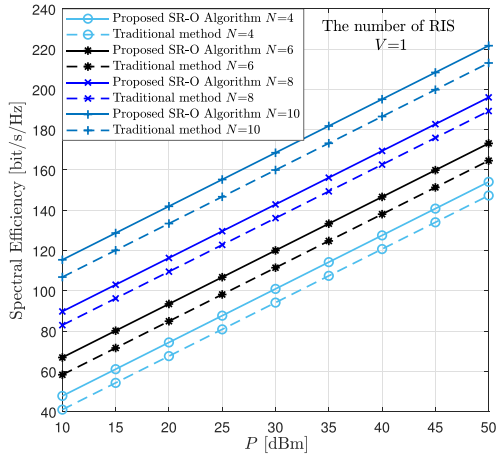

Fig. 14. Spectral efficiency of the proposed SR-O method.

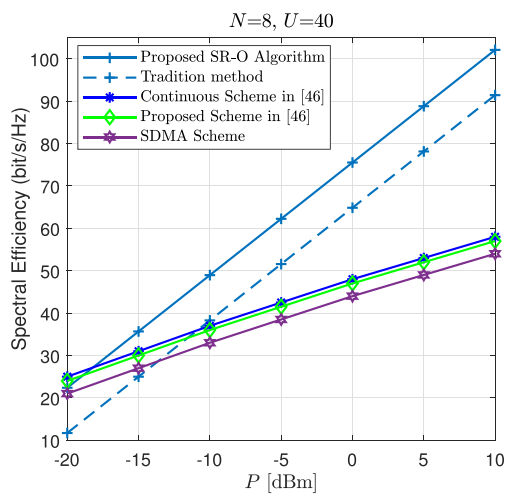

Fig. 15. Comparison of spectral efficiency with other current advanced algorithms for RIS-RSMA scheme.

through user communication time allocation where the coefficients were set according to the type of data transmitted by users. Due to the highly coupled nature of the proposed optimization problem, it was divided into two sub-problems: one involving time allocation coefficients and the other addressing power allocation under constraints based on training symbols and data symbols. This method demonstrated that RIS in RSMA systems can offer more potential and innovative approaches for efficient resource utilization, providing robust support for the development of IoT network.

Appendix A
Proof of the Upper Bound of SNR

According to optimal power distribution in a training-based scheme [49], the optimal power allocation can be expressed as:

$$\omega_{op} = \begin{cases} \Lambda - \sqrt{\Lambda(\Lambda-1)}, M_{u,d} > N, \\ \dfrac{1}{2}, \qquad\qquad M_{u,d} = N, \\ \Lambda + \sqrt{\Lambda(\Lambda-1)}, M_{u,d} < N, \end{cases} \tag{48}$$

where $\Lambda = \frac{N+\Upsilon M_u}{\Upsilon M_u(1-\frac{N}{M_{u,d}})}$, $\Upsilon T = \Upsilon_t T_t + \Upsilon_d T_d$, which T_t and T_d denote time of training symbols and transmitting data symbols, Υ_t and Υ_d express SNR of the two phase, Υ is the expected received SNR at each receive antenna. In order to maximize Υ_{up} over condition $0 < \omega_{op} < 1$, consider the following three cases:

(1) $M_{u,d} = N$, $\Upsilon_{up} = \frac{(\Upsilon M_u)^2}{M_u(M_u+\Upsilon M_u)}\omega_{op}(1-\omega_{op})$, follows $\omega_{op} = \dfrac{1}{2}$, therefore we can get $\Upsilon_{up} = \dfrac{(\Upsilon M_u)^2}{4M_u(M_u+\Upsilon M_u)}$.

(2) $M_{u,d} > N$, $\Upsilon_{up} = \dfrac{\Upsilon M_u}{M_{u,d}-M_u} \cdot \dfrac{\omega_{op}(1-\omega_{op})}{-\omega_{op}+\Lambda}$ when $\Lambda > 1$, because $\arg\max_{0<\omega_{op}<1} \dfrac{\omega_{op}(1-\omega_{op})}{-\omega_{op}+\Lambda} = \Lambda - \sqrt{\Lambda(\Lambda-1)}$, the result is obtained as $\Upsilon_{up} = \frac{\Upsilon M_u}{M_{u,d}-M_u} \cdot (\sqrt{\Lambda} - \sqrt{\Lambda-1})^2$.

(3) $M_{u,d} < N$, $\Upsilon_{vp} = \dfrac{\Upsilon M_u}{M_u-M_{u,d}} \cdot \dfrac{\omega_{op}(1-\omega_{op})}{\omega_{op}-\Lambda}$ when $\Lambda < 0$, because $\arg\max_{0<\omega_{op}<1} \dfrac{\omega_{op}(1-\omega_{op})}{\omega_{op}-\Lambda} = \Lambda + \sqrt{\Lambda(\Lambda-1)}$, the result is obtained as $\Upsilon_{up} = \dfrac{\Upsilon M_u}{M_u-M_{u,d}} \cdot (\sqrt{-\Lambda} - \sqrt{-\Lambda+1})^2$.

Appendix B
Proof of Optimal RIS Phase Shifts

The formula for the channel gain maximization problem for the distant l-th user under RIS assistance is given as:

$$\max_{\theta_{v,1},\ldots,\theta_{v,z}} \left|\mathbf{h}_l + \tilde{\mathbf{h}}_{vz}^l \Theta_{v,z} \tilde{\mathbf{H}}_{vz}\right|^2. \tag{49}$$

Assume that all RIS elements have the same amplitude gain. The matrix multiplication in (48) can be extended as:

$$\max_{\theta_{v,1},\ldots,\theta_{v,z}} \left|\mathbf{h}_l + g_{v,z} \sum_{z=1}^{Z} \left(\left[\tilde{\mathbf{H}}_{vz}\right]_z \left[\tilde{\mathbf{h}}_{vz}^l\right]_z e^{j\theta_{v,z}}\right)\right|^2, \tag{50}$$

where $[\mathbf{h}]_z$ represents the channel gain corresponding to the z-th element. Using the triangle inequality, i.e., $|\sum_{i=1}^n a_i| \leq \sum_{i=1}^n |a_i|$, equation (50) can be expressed as:

$$\left|\mathbf{h}_l + g_{v,z} \sum_{z=1}^{Z} \left(\left[\tilde{\mathbf{H}}_{vz}\right]_z \left[\tilde{\mathbf{h}}_{vz}^l\right]_z e^{j\theta_{v,z}}\right)\right|^2$$
$$\leq \left(|\mathbf{h}_l| + \left|\mathbf{g}_{v,z} \sum_{z=1}^{Z} \left(\left[\tilde{\mathbf{H}}_{vz}\right]_z\right]_z \left[\tilde{\mathbf{h}}_{vz}^l\right]_z e^{j\theta_{v,z}}\right|\right)^2. \tag{51}$$

The following conditions should be satisfied to make the inequality sign in (51) become the equal sign:

$$\arg(\mathbf{h}_l) = \arg\left(g_{v,z} \sum_{z=1}^{Z} \left(\left[\tilde{\mathbf{H}}_{vz}\right]_z \left[\tilde{\mathbf{h}}_{vz}^l\right]_z e^{j\theta_{v,z}}\right)\right). \tag{52}$$

The upper bound of (51) is obtained using the triangle inequality in summation over v:

$$|\mathbf{h}_l| + \left|g_{v,z} \sum_{z=1}^{Z} \left(\left[\tilde{\mathbf{H}}_{vz}\right]_z \left[\tilde{\mathbf{h}}_{vz}^l\right]_z e^{j\theta_{v,z}}\right)\right|$$

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$$\leq |\mathbf{h}_l| + g_{v,z} \sum_{z=1}^{Z} \left(\left| \tilde{\mathbf{H}}_{vz} \right|_z \left| \tilde{\mathbf{h}}_{vz}^l \right|_z \left| e^{j\theta_{v,z}} \right| \right). \tag{53}$$

The condition that the upper bound can be achieved is:

$$\arg\left(\left[\tilde{\mathbf{H}}_{vz} \right]_z \left[\tilde{\mathbf{h}}_{vz}^l \right]_z e^{j\theta_{v,z}} \right) = \arg\left(\left[\tilde{\mathbf{H}}_{vz} \right]_{z+1} \left[\tilde{\mathbf{h}}_{vz}^l \right]_{z+1} e^{j\theta_{v,z+1}} \right). \tag{54}$$

(54) means that the signal reflections received by UE from Z different RIS elements are in phase, so the summation can be ignored in (53) and rewritten as:

$$\arg(\mathbf{h}_l) = \arg\left(\left[\tilde{\mathbf{H}}_{vz} \right]_z \left[\tilde{\mathbf{h}}_{vz}^l \right]_z e^{j\theta_{v,z}} \right). \tag{55}$$

At this point, the reflected signal received by the UE is in phase with the signal transmitted by the BS, which means that all reflected signals and direct-link signals will undergo constructive interference at the UE, thereby enhancing the signal strength. The optimal phase shift is obtained as:

$$\theta_{v,z}^* = \arg(\mathbf{h}_l) - \arg\left(\left[\tilde{\mathbf{H}}_{vz} \right]_z \left[\tilde{\mathbf{h}}_{vz}^l \right]_z \right). \tag{56}$$

Similarly, the optimal RIS phase shift for the signal received by user K can be obtained. Subsequently, the optimal phase shift $\theta_{v,z}^*$ is used as the RIS configuration for power allocation.

Appendix C
Proof of the Optimal Power Distribution of User

The rate is weighted and maximized by allocating power reasonably to each user. The problem formula is expressed as follows:

$$\max_{P_1,\ldots,P_u} \sum_{u=1}^{U} \alpha_u R_{u,\max}, \quad \text{s.t.} \sum_{*=c,p} \sum_{u=1}^{U} P_{*,u} = P, P_u \geq 0. \tag{57}$$

Let $\sum_{u=1}^{U} \frac{(p_{h,ave}^2 - \mathcal{M}_e)\beta_{ii}\mu_{ii}}{\frac{P_d}{M_{u,d}}\mathcal{M}_e + \sigma_n^2} = \sum_{u=1}^{U} P_u \cdot \frac{1}{\tilde{L}_u}$, which $\tilde{L}_u = \frac{(p_{h,ave}^2 - \mathcal{M}_e)\beta_{ii}\mu_{ii}}{(\frac{P_d}{M_{u,d}}\mathcal{M}_e + \sigma_n^2)P_u}$, then the objective function is reexpressed as:

$$\begin{aligned} \sum_{u=1}^{U} \alpha_u R_{u,\max} &= \sum_{\Delta=K,L} \alpha_\Delta R_{\Delta,\max} \\ &= \sum_{\Delta=K,L} \alpha_\Delta \log\left(1 + P_\Delta \cdot \frac{1}{\tilde{L}_\Delta}\right) \\ &= \sum_{\Delta=K,L} \alpha_\Delta \log(\tilde{L}_\Delta + P_\Delta) \\ &\quad + \sum_{\Delta=K,L} \alpha_\Delta \log \frac{1}{\tilde{L}_\Delta}. \end{aligned} \tag{58}$$

The optimization problem (57) then change into:

$$\max_{P_\Delta} \sum_{\Delta=K,L} \alpha_\Delta \log(\tilde{L}_\Delta + P_\Delta), \quad s.t. \sum_{*=c,p} \sum_{\Delta=K,L} P_{*,\Delta} = P, P_{*,\Delta} \geq 0. \tag{59}$$

This problem is a convex optimization problem, and satisfies the Slater condition. In this case, Karush-Kuhn-Tucker (KKT) condition is taken as the supplementary condition of optimality, λ and μ' are introduced as Lagrange multipliers, and the Lagrange function is:

$$\begin{aligned} \tilde{L} = &- \sum_{\Delta=K,L} \alpha_\Delta \log\left(\tilde{L}_\Delta + P_\Delta\right) \\ &- \sum_{\Delta=K,L} \lambda_\Delta P_\Delta + \mu' \left[\left(\sum_{*=c,p} \sum_{\Delta=K,L} P_{*,\Delta} \right) - P \right]. \end{aligned} \tag{60}$$

The complementary relaxation condition is $\lambda_\Delta P_\Delta = 0$, the stationarity condition is $-\frac{\alpha_\Delta}{\tilde{L}_\Delta + P_\Delta} - \lambda_\Delta + \mu\prime = 0$, thus get $(\mu\prime - \frac{\alpha_\Delta}{\tilde{L}_\Delta + P_\Delta})P_\Delta = 0$. When $\mu' \geq \frac{\alpha_\Delta}{\tilde{L}_\Delta}$, there is $P_\Delta = 0$, when $\mu' < \frac{\alpha_\Delta}{\tilde{L}_\Delta}$, $P_\Delta = \frac{\alpha_\Delta}{\mu'} - \tilde{L}_\Delta$.

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